Safe System of Work
Undertaking Vehicle Tyre Repairs

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O v e r v i e w

Explosive energy

Inflated tyres contain a large amount of stored energy. For example, the sidewall of a typical commercial vehicle (CV) tyre has over 34 tonnes of force acting on it. Tyres are designed to withstand this but if they are damaged or used while flat, or significantly underinflated, they may fail. The force can then be released explosively at an angle of up to 45 degrees from the rupture (which is often, but not always, the face of the sidewall), resulting in a destructive air blast and the ejection of high-speed particles. If the wheel is not restrained, it can fly metres through the air.

Personal Protective Equipment (PPE) Requirements







Safe Workplaces

The information below specifically deals with the Risks relating to tyre inflation. WF024 Tyre Maintenance and WF026 General Jacking and Blocking in the Fox Group Wagon Fitters Handbook (FG1.0) cover all Risks associated with Tyre Repairs and the necessary Control Measures.
Risks Associated with Tyre Repairs include:
■ manual handling of the tyre and wheel;
■ collapse of an elevated vehicle;
■ being struck by other vehicles (roadside);
■ explosion of the tyre or disintegration of the wheel during inflation. Risk of failure During Inflation
There will be an increased risk of failure:

■ following tyre repair;
■ where there has been sidewall damage (e.g. after a tyre has been run flat or significantly underinflated);
■ when fitting a tyre to a split-rim wheel.

Safety during tyre inflation

Cuts and splits should be clearly visible by inspection and there are industry standards on whether a repair should be attempted. However, damage to the internal steel or textile cords may not be obvious. This can be caused by impact (e.g. hitting a kerb or pothole) or if the tyre is run significantly underinflated (normally taken as below 80% of its recommended pressure). When significantly underinflated, the tyre's sidewall flexes excessively and the cords become increasingly damaged the longer the wheel is run. Also, deflation may not be detected for some time, particularly on multi-wheel axles, where the exterior tyre is undamaged.
This internal damage may not become obvious until the tyre is reinflated and a bulge occurs. At this stage, the additional strain placed on the adjacent cords can cause them to break in rapid succession, spreading around the sidewall until the casing splits apart violently. This is commonly known as a 'zipper-failure'. To reduce this risk:
■ Before deflating a tyre, check the pressure and chalk the reading on the tyre wall. Remember, low tyre pressure may have caused tyre wall damage.
■ Do not inflate any tyre that has been significantly underinflated until it has been adequately checked. Examine wheels and tyres (externally and internally) for signs of damage, e.g. cracks, 'marbling' (black lines), bulging, soft spots or exposed steel cord in the tyre carcass. If in doubt, do not reinflate the tyre.
■ When reinflating, take particular care to stay outside the likely explosion trajectory. Watch and listen for signs that might indicate a zipper failure. If you suspect a problem, do not approach the tyre to deflate it – use the quick-release connection at the operator's end of the hose.
■ Use a clip-on chuck to connect the airline with a quick release coupling at the operator's end (this allows tyre deflation from a safe position if problems occur).
■ Use airline hoses long enough to allow the operator to stay outside the likely explosion trajectory during inflation.
■ Use enough bead lubricant when seating the tyre. Consider removing the valve core or using a 'beadblaster' if seating is difficult.
■ Remove the airline after use to prevent air seepage and possible over inflation.
■ Don't use valve connectors that require the operator to hold them in place.
■ Don't exceed the manufacturer's recommended tyre pressure for the size and rating of the tyre.
■ Don't use 'unrestricted' airlines (ie without a gauge or pressure control device).
■ Don't allow the control valve to be jammed open (which could allow the operator to leave the inflating tyre unattended).

Tyres on commercial vehicles

Extra safety measures are needed for inflating larger, well-based (ie single-piece) CV tyres above 15 psi. This includes some light CV tyres which are inflated to around 70 psi and may cause serious injury in the event of failure. The extra measures include using a restraining device such as:
■ a strong, firmly secured tyre inflation cage. Consider lining this with mesh to retain debris. For fixed installations, it is helpful to mark the safety exclusion zone on the workshop floor as a reminder to staff;
■ a secured horizontal stool and associated clamping mechanism;
■ a portable restraint. Examples include a lightweight cover that encloses the tyre and wheel rim. This

may be particularly useful for off-site repairs. In the event of an explosion the fabric contains projected debris. Fabric devices may need to be replaced after an explosion.